

Gaussian covariance of windowed bispectrum multipoles in the tripolar (Sugiyama) basis

The same formalism as for the power spectrum: local plane-parallel correlators, plane-wave and Legendre expansions, Wigner–Eckart, and a window function from pair counts

Derivation note, branch thecov2, October 2026. Conventions follow Sugiyama, Saito, Beutler & Seo (2019, 2020).

Summary. The Gaussian (*PPP*) covariance of the bispectrum multipoles $B_{\ell_1 \ell_2 L}(k_1, k_2)$ of a windowed survey can be derived exactly as the power-spectrum covariance was: each of the three two-point correlators that a Wick pairing produces is a windowed, locally anisotropic power spectrum; the shell integrals turn into Bessel averages; the Legendre weights of the estimators and of the anisotropic correlators turn into spherical harmonics; and the Wigner–Eckart theorem collapses the product of Gaunt coefficients onto rotational invariants. Two things are new. First, the three correlators sit at three points, so the window enters through its **three-point** function $\int d^3x W(x)W(x+r_1)W(x+r_2)$, weighted by an invariant combination of harmonics of the two separations $\hat{r}_1, \hat{r}_2$ and of the three lines of sight. Second, the third side of the triangle, $k_3 = |k_1 + k_2|$, is not binned, so its power spectrum depends on the opening angle of the two binned momenta; this couples the two separations through a bipolar expansion. The six pairings split into two classes: in the two pairings where the two line-of-sight galaxies pair with each other the window function is anchored at that galaxy and is a product of two pair sums per anchor — no triplet counting; in the other four pairings one shell momentum runs along the third side of the triangle. For the elements that matter most, the diagonal $(k_1, k_2) = (k'_1, k'_2)$, the first class is the leading term and the second is suppressed by $\sim \Delta k/k$. In a periodic box the result reduces to eq. (49) of Sugiyama et al. (2020). Nothing here is specific to B_{000} or B_{202} ; their index sets are listed at the end.

Status (October 2026). The Gaussian part was reviewed by independent numerical checks; the derivation stands, with these corrections not yet folded into the text: (i) $H_{202} = +1/\sqrt{5}$ (in general $H_{\ell 0 \ell} = (-1)^\ell/\sqrt{2\ell+1}$); (ii) the invariant basis $E_{K_1 K_2}$ after Equation 25 must couple the line-of-sight harmonics with the same variance as T : write $\bar{Y}_{LM} = (-1)^M Y_{L, -M}$, so that all five harmonics are unconjugated, and use $E = \sum \langle \lambda_1 \rho_1 L_1 M_1 | K_1 \kappa_1 \rangle \langle \lambda_2 \rho_2 L_2 M_2 | K_2 \kappa_2 \rangle \begin{pmatrix} K_1 & K_2 & \Lambda_3 \\ \kappa_1 & \kappa_2 & \mu_3 \end{pmatrix}$, which is orthonormal ($\mathcal{T} = \sum TE$) and makes the scalar of Equation 27 and the anchored moments $A^{(a)}$ rotation invariant; the same correction applies to the pair-sum formulae; (iii) in a periodic box only $\lambda_1 = \lambda_2 = 0$ survive, and $\mathcal{Q} \rightarrow V\sqrt{N_{L_1 L_2 \Lambda_3}}/4\pi H_{L_1 L_2 \Lambda_3}$ carries the redshift-space ranks L_a, Λ_3 of eq. (49); (iv) in class 2 the window does not bound the (λ, λ') sums in the (p_1, p_2) variables; use $(p_1, p_3 = p_1 + p_2)$ so that each anchored side carries one plane wave, and the ranks are bounded by $\lambda_w + \ell_a + \ell'_a + L_a$ as in class 1 (the class-1 (a, b) bound also reads so); (v) the pairing $(12)(1'2')(33')$ is not negligible: on the elements with $i = j, i' = j'$ it has the same coincidence measure as class 1 and is suppressed only by the integral constraint, $C^{(D)}(ii; i'i') \approx (4\pi)^3 H H' \sqrt{N N'} / (I_3^2 V_i V_{i'}) \int d^3x d^3x' W(x)W(x') \langle \tilde{S} P \rangle_i \langle \tilde{S}' P \rangle_{i'} \langle F(x)F(x') \rangle_{\text{IC}}$, the long-mode variance of the W -weighted field (zero for uniform ω , tens of percent of the (i, i) variance in a Gaussian test with a radially varying ω); (vi) the self-pair $y_1 = y_2$ must be subtracted from the $r_1 = r_2$ bins of the pair sums, and the product cost is $\sim 10^{12}$ with $\mu_3 = 0$, not 10^{11} .

1 Conventions

We use the identities of the power-spectrum note: the addition theorem, the plane-wave expansion, the generalised Gaunt coefficient $G_{\ell_1 \dots \ell_n}^{m_1 \dots m_n} \equiv \int d\Omega \prod_a Y_{\ell_a m_a}$, and the product rule

$$\prod_{a=1}^n Y_{\ell_a m_a}(\hat{\mathbf{n}}) = \sum_{\Lambda\mu} G_{\ell_1 \dots \ell_n \Lambda}^{m_1 \dots m_n \mu} \bar{Y}_{\Lambda\mu}(\hat{\mathbf{n}}), \quad \prod_{a=1}^n \bar{Y}_{\ell_a m_a}(\hat{\mathbf{n}}) = \sum_{\Lambda\mu} G_{\ell_1 \dots \ell_n \Lambda}^{m_1 \dots m_n \mu} Y_{\Lambda\mu}(\hat{\mathbf{n}}). \quad (1)$$

Spherical harmonics are the standard $Y_{\ell m}$; Sugiyama et al. use $y_\ell^m = \sqrt{4\pi/(2\ell+1)} Y_{\ell m}$.

Fields and windows. One tracer, weighted field $F(x) = w(x)[n_g(x) - \alpha n_r(x)]$, Fourier transform $F(k) = \int d^3x e^{-ik \cdot x} F(x)$, so $F(-k) = \overline{F(k)}$. With $\omega(x) \equiv \bar{n}(x)w(x)$:

$$W(x) \equiv \omega^2(x) \quad (\text{pair window}), \quad S(x) \equiv (1 + \alpha) \bar{n}(x) \omega^2(x) \quad (\text{shot-noise window}), \quad (2)$$

$$I_3 \equiv \int d^3x \omega^3(x).$$

I_3 is the normalisation $I = \int d^3x \bar{n}^3 w^3$ of Sugiyama et al. (2019, eq. 37).

Shell averages. For a shell i of the wavenumber magnitude, $V_i = \int_i d^3q/(2\pi)^3$ and

$$\langle\langle f \rangle\rangle_i \equiv \frac{1}{V_i} \int_i \frac{d^3q}{(2\pi)^3} f(q) = \frac{\int_i q^2 dq \int \frac{d\Omega_q}{4\pi} f(q)}{\int_i q^2 dq}, \quad (3)$$

so that the angular part is a normalised average; nested averages are abbreviated, $\langle\langle f \rangle\rangle_{ij} \equiv \langle\langle\langle f \rangle\rangle_i\rangle_j$ and $\langle\langle f \rangle\rangle_{ijij'}$ for the four shells of a covariance element. As in the power-spectrum note, with P_L the multipoles of the model power spectrum,

$$\bar{j}_\lambda^{(i)}(r) \equiv \frac{\int_i q^2 dq j_\lambda(qr)}{\int_i q^2 dq}, \quad P_{L\lambda}^{(i)}(r) \equiv \frac{\int_i q^2 dq j_\lambda(qr) P_L(q)}{\int_i q^2 dq}. \quad (4)$$

Plane-wave shell averages. From the plane-wave expansion and the orthogonality of the harmonics,

$$\langle\langle e^{-iq \cdot r} \bar{Y}_{\ell m}(\hat{q}) \rangle\rangle_i = (-i)^\ell \bar{j}_\ell^{(i)}(r) \bar{Y}_{\ell m}(\hat{r}), \quad \langle\langle e^{+ip \cdot r} \bar{Y}_{\Lambda\mu}(\hat{p}) g(p) \rangle\rangle_i = i^\Lambda \bar{g}_\Lambda^{(i)}(r) \bar{Y}_{\Lambda\mu}(\hat{r}), \quad (5)$$

with $\bar{g}_\Lambda$ the Bessel-weighted radial average of g . The first is used for the momenta of the estimator whose covariance we take, the second for the primed one.

Tripolar basis and estimator (Sugiyama et al. 2019, eqs. 12–13 and 34–35). With $H_{\ell_1 \ell_2 L} = \begin{pmatrix} \ell_1 & \ell_2 & L \\ 0 & 0 & 0 \end{pmatrix}$ and $N_{\ell_1 \ell_2 L} = (2\ell_1 + 1)(2\ell_2 + 1)(2L + 1)$,

$$B(k_1, k_2, \hat{\mathbf{n}}) = \sum_{\ell_1 \ell_2 L} B_{\ell_1 \ell_2 L}(k_1, k_2) \mathcal{S}_{\ell_1 \ell_2 L}(\hat{k}_1, \hat{k}_2, \hat{\mathbf{n}}), \quad (6)$$

$$\mathcal{S}_{\ell_1 \ell_2 L}(\hat{a}, \hat{b}, \hat{\mathbf{n}}) = \frac{1}{H_{\ell_1 \ell_2 L}} \sum_{m_1 m_2 M} \begin{pmatrix} \ell_1 & \ell_2 & L \\ m_1 & m_2 & M \end{pmatrix} y_{\ell_1}^{m_1}(\hat{a}) y_{\ell_2}^{m_2}(\hat{b}) y_L^M(\hat{\mathbf{n}}),$$

$$B_{\ell_1 \ell_2 L}(k_1, k_2) = N_{\ell_1 \ell_2 L} H_{\ell_1 \ell_2 L}^2 \int \frac{d\Omega_1 d\Omega_2 d\Omega_n}{(4\pi)^3} \bar{\mathcal{S}}_{\ell_1 \ell_2 L} B. \quad (7)$$

It is convenient to write $\mathcal{S}$ through the unnormalised combination of ordinary harmonics,

$$\tilde{\mathcal{S}}_{\ell_1 \ell_2 L}(\hat{a}, \hat{b}, \hat{\mathbf{n}}) \equiv \sum_{m_1 m_2 M} \begin{pmatrix} \ell_1 & \ell_2 & L \\ m_1 & m_2 & M \end{pmatrix} Y_{\ell_1 m_1}(\hat{a}) Y_{\ell_2 m_2}(\hat{b}) Y_{LM}(\hat{\mathbf{n}}), \quad \mathcal{S} = \frac{(4\pi)^{3/2}}{H\sqrt{N}} \tilde{\mathcal{S}}. \quad (8)$$

The estimator attaches the line of sight to the third galaxy, which is the one not integrated over a shell:

$$\hat{B}_{\ell_1 \ell_2 L}(i, j) = \frac{NH^2}{I_3} \left\langle \left\langle \int d^3x \bar{\mathcal{S}}_{\ell_1 \ell_2 L}(\hat{q}_1, \hat{q}_2, \hat{\mathbf{x}}) e^{i(q_1+q_2) \cdot x} F(q_1) F(q_2) F(x) \right\rangle \right\rangle_{ij} \quad (9)$$

$$= \frac{(4\pi)^{3/2} H \sqrt{N}}{I_3} \left\langle \left\langle \int d^3x \bar{\tilde{\mathcal{S}}}_{\ell_1 \ell_2 L}(\hat{q}_1, \hat{q}_2, \hat{\mathbf{x}}) e^{i(q_1+q_2) \cdot x} F(q_1) F(q_2) F(x) \right\rangle \right\rangle_{ij}.$$

Written out, $e^{i(q_1+q_2)\cdot x} F(q_1)F(q_2)F(x) = \int d^3y_1 d^3y_2 F(y_1)F(y_2)F(x) e^{-iq_1\cdot(y_1-x)} e^{-iq_2\cdot(y_2-x)}$: the two shell momenta are the sides of the triangle from the line-of-sight galaxy at x to the galaxies at y_1, y_2 .

Unbiasedness. In the local approximation $\langle F(y_1)F(y_2)F(x) \rangle \approx \omega^3(x) \zeta(y_1 - x, y_2 - x; \hat{x})$, so $\langle \hat{B} \rangle = (NH^2/I_3) \int d^3x \omega^3 \langle \int d\Omega_1 d\Omega_2 / (4\pi)^2 \bar{\mathcal{S}}(\hat{q}_1, \hat{q}_2, \hat{x}) B(q_1, q_2, \hat{x}) \rangle$, which is $B_{\ell_1 \ell_2 L}$ averaged over the shells and over the survey, because for fixed $\hat{n}$ the $(\hat{q}_1, \hat{q}_2)$ average of $\bar{\mathcal{S}}_{\ell_1 \ell_2 L} \mathcal{S}_{\ell'_1 \ell'_2 L'}$ is $\delta_{\ell_1 \ell'_1} \delta_{\ell_2 \ell'_2} \delta_{LL'}/(NH^2)$ (orthogonality of the 3- j symbols at fixed M). The window dilutes it by $\int \omega^3/I_3 = 1$; with a pair-averaged window this ratio is the bispectrum analogue of I_k/norm .

2 Wick expansion

Write the covariance as $\langle \hat{B} \bar{\hat{B}}' \rangle_c$ with the primed estimator at shells (i', j') and multipoles $(\ell'_1 \ell'_2 L')$. Since $\overline{F(p)} = F(-p)$,

$$\overline{\hat{B}}'_{\ell'_1 \ell'_2 L'}(i', j') = \frac{(4\pi)^{3/2} H' \sqrt{N'}}{I_3} \left\langle \int d^3x' \tilde{\mathcal{S}}_{\ell'_1 \ell'_2 L'}(\hat{p}_1, \hat{p}_2, \hat{x}') e^{-i(p_1+p_2)\cdot x'} F(-p_1)F(-p_2)F(x') \right\rangle_{i' j'} \quad (10)$$

Label the six fields $1 \equiv F(q_1)$, $2 \equiv F(q_2)$, $3 \equiv F(x)$, $1' \equiv F(-p_1)$, $2' \equiv F(-p_2)$, $3' \equiv F(x')$. The Gaussian part of the six-point function is a sum over pairings; those with a pair inside one estimator, e.g. $\langle F(q_1)F(q_2) \rangle$, are proportional to the window at $q_1 + q_2$ and only survive when $|q_1 + q_2|$ is below the window scale, a fraction $\sim (\Delta k/k)^2$ of the shells: we drop them, as the power-spectrum note drops the $k \approx 0$ terms. What remains are the $3! = 6$ pairings between $\{1, 2, 3\}$ and $\{1', 2', 3'\}$:

$$\begin{aligned} \text{class 1: } & (11')(22')(33'), \quad (12')(21')(33'); \\ \text{class 2: } & (11')(23')(32'), \quad (12')(23')(31'), \quad (13')(21')(32'), \quad (13')(22')(31'). \end{aligned} \quad (11)$$

In class 1 the two line-of-sight galaxies pair with each other; in class 2 each pairs with a shell galaxy of the other estimator.

3 The correlators in the local approximation

Three kinds of two-point correlator occur. With $P(p; \hat{n}) = \sum_L \mathcal{L}_L(\hat{p} \cdot \hat{n}) P_L(p)$, a pair of Fourier fields,

$$\langle F(q)F(-p) \rangle = \int d^3y d^3y' e^{-iq\cdot y + ip\cdot y'} \langle F(y)F(y') \rangle \approx \sum_L \int d^3y e^{-i(q-p)\cdot y} W(y) \mathcal{L}_L(\hat{p} \cdot \hat{y}) P_L(p), \quad (12)$$

exactly as in the power-spectrum note (window at y , anisotropy about $\hat{y}$, pair separation only in the Fourier kernel; the power spectrum is evaluated at the primed momentum). A Fourier field and a configuration-space field, whose position carries the phase $e^{-ip_3\cdot x'}$ of the primed estimator and is integrated,

$$\int d^3x' e^{-ip_3\cdot x'} \langle F(q)F(x') \rangle \approx \int d^3y e^{-i(q+p_3)\cdot y} W(y) P(p_3; \hat{y}), \quad (13)$$

obtained by writing $x' = y + u$ and $\int d^3u e^{-ip_3\cdot u} \xi(u; \hat{y}) = P(p_3; \hat{y})$; and the two configuration-space fields, with both estimator phases,

$$\int d^3x' e^{i(q_1+q_2)\cdot x - i(p_1+p_2)\cdot x'} \langle F(x)F(x') \rangle \approx W(x) e^{i(q_1+q_2-p_1-p_2)\cdot x} P(p_1 + p_2; \hat{x}). \quad (14)$$

In every case the line-of-sight harmonic of the primed estimator, $Y_{L'M'}(\hat{x}')$, is moved to the position of the galaxy its partner sits at ($\hat{x}$ in Equation 14, $\hat{y}$ in Equation 13), as the Legendre weight of the estimator was moved in the power-spectrum note. Shot noise adds to each correlator the same expression with $W \rightarrow S$ and $P_L \rightarrow \delta_{L0}^K$; we restore it at the end.

4 Class 1: the two lines of sight pair together

4.1 Phases and the three-point window function

For the pairing (11')(22')(33') the three correlators are Equation 12 for (q_1, p_1) at y_1 , Equation 12 for (q_2, p_2) at y_2 , and Equation 14 at x . The phases combine into

$$e^{-i(q_1-p_1)\cdot y_1} e^{-i(q_2-p_2)\cdot y_2} e^{i(q_1+q_2-p_1-p_2)\cdot x} = e^{-i(q_1-p_1)\cdot r_1} e^{-i(q_2-p_2)\cdot r_2}, \quad r_1 \equiv y_1 - x, \quad r_2 \equiv y_2 - x : (15)$$

each momentum pair rides a plane wave on one side of the triangle from the line-of-sight galaxy. Collecting everything,

$$C_{\ell\ell'}^{(1)}(ij; i'j') = \frac{(4\pi)^3 HH' \sqrt{NN'}}{I_3^2} \langle \langle \mathcal{I}^{(1)} \rangle \rangle_{ij i'j'}, \quad \mathcal{I}^{(1)} \equiv \int d^3x d^3r_1 d^3r_2 W(x) W(x+r_1) W(x+r_2) \quad (16)$$

$$\times e^{-i(q_1-p_1)\cdot r_1} e^{-i(q_2-p_2)\cdot r_2} \overline{\tilde{\mathcal{S}}_{\ell_1 \ell_2 L}(\hat{q}_1, \hat{q}_2, \hat{x})} \tilde{\mathcal{S}}_{\ell'_1 \ell'_2 L'}(\hat{p}_1, \hat{p}_2, \hat{x}) P(p_1; \hat{y}_1) P(p_2; \hat{y}_2) P(p_1 + p_2; \hat{x}),$$

with $\hat{y}_a = (x + r_a)/|x + r_a|$. This is the master formula of class 1. The second class-1 pairing, (12')(21')(33'), is the same expression with the primed labels exchanged, $(p_1, \ell'_1, i') \leftrightarrow (p_2, \ell'_2, j')$; for the diagonal elements with $i = j$ it adds to the first with equal weight.

Box limit. For a periodic box of volume V with a global line of sight, $W = 1$, $\hat{x} = \hat{n}$, $I_3 = V$, and $\int d^3r_1 e^{-i(q_1-p_1)\cdot r_1} = (2\pi)^3 \delta_D(q_1 - p_1)$, whose shell average over p_1 is $\delta_{ii'}/V_i$. Hence

$$C_{\text{box}}^{(1)} = \frac{(4\pi)^3 HH' \sqrt{NN'}}{V V_i V_j} \delta_{ii'} \delta_{jj'} \left\langle \langle \overline{\tilde{\mathcal{S}}_{\ell} \tilde{\mathcal{S}}_{\ell'}} P(q_1) P(q_2) P(|q_1 + q_2|) \rangle \right\rangle_{ij} \quad (17)$$

$$= \frac{NN' H^2 H'^2 V}{\tilde{N}_{\text{mode}(k_i)} \tilde{N}_{\text{mode}(k_j)}} \delta_{ii'} \delta_{jj'} \langle \langle \mathcal{S}_{\ell} \mathcal{S}_{\ell'} PPP \rangle \rangle_{ij},$$

using $(4\pi)^3 \overline{\tilde{\mathcal{S}} \tilde{\mathcal{S}'}} = HH' \sqrt{NN'} \mathcal{S} \mathcal{S}'$ and $V_i = \tilde{N}_{\text{mode}(k_i)}/V$ with $\tilde{N}_{\text{mode}} = 4\pi k^2 \Delta k V / (2\pi)^3$. This is the first term of eq. (49) of Sugiyama et al. (2020), with their top-hat $W(k, k')$ as $\delta_{ii'}$; the “5 perms.” there are the other five pairings. With shot noise restored, $P \rightarrow P + 1/\bar{n}$ in each factor, as in their $P^{(N)}$.

4.2 Harmonic expansion

Momenta of the unprimed estimator. $\overline{\tilde{\mathcal{S}}_{\ell}}(\hat{q}_1, \hat{q}_2, \hat{x})$ carries $\overline{Y}_{\ell_1 m_1}(\hat{q}_1) \overline{Y}_{\ell_2 m_2}(\hat{q}_2) \overline{Y}_{LM}(\hat{x})$ and nothing else depends on $\hat{q}_1, \hat{q}_2$, so by Equation 5

$$\langle \langle e^{-i q_1 \cdot r_1} \overline{Y}_{\ell_1 m_1}(\hat{q}_1) \rangle \rangle_i = (-i)^{\ell_1} \bar{j}_{\ell_1}^{(i)}(r_1) \overline{Y}_{\ell_1 m_1}(\hat{r}_1), \quad (18)$$

and likewise for q_2 . The unprimed estimator contributes only bin-averaged Bessel functions, with no power spectrum, as in the power-spectrum note.

The vertex term. The third correlator depends on $\hat{p}_1$ and $\hat{p}_2$ jointly, through $p_3 = p_1 + p_2$:

$$P(p_1 + p_2; \hat{x}) = \sum_{L_3} \frac{4\pi}{2L_3 + 1} \sum_{\gamma} \overline{Y}_{L_3 \gamma}(\hat{x}) [Y_{L_3 \gamma}(\hat{p}_3) P_{L_3}(|p_1 + p_2|)]. \quad (19)$$

The bracket is a function of $(\hat{p}_1, \hat{p}_2)$ at fixed magnitudes that transforms as a rank- L_3 tensor; its bipolar expansion

$$Y_{L_3 \gamma}(\hat{p}_3) P_{L_3}(|p_1 + p_2|) = \sum_{ab} c_{ab}^{L_3}(p_1, p_2) [Y_a(\hat{p}_1) \otimes Y_b(\hat{p}_2)]_{L_3 \gamma}, \quad (20)$$

$$[Y_a \otimes Y_b]_{L \gamma} \equiv \sum_{\alpha \beta} \langle a \alpha b \beta | L \gamma \rangle Y_{a \alpha}(\hat{p}_1) Y_{b \beta}(\hat{p}_2),$$

has coefficients $c_{ab}^{L_3}(p_1, p_2) = \int d\Omega_1 d\Omega_2 \overline{[Y_a \otimes Y_b]_{L_3\gamma}} Y_{L_3\gamma}(\hat{p}_3) P_{L_3}(|p_1 + p_2|)$ (independent of γ), a one-dimensional integral over the opening angle after the rotational symmetry is used. Parity gives $a + b$ even, and $|a - b| \leq L_3 \leq a + b$. For $L_3 = 0$ the coefficients are the Legendre coefficients of $P_0(|p_1 + p_2|)$ in $\hat{p}_1 \cdot \hat{p}_2$: $a = b$ and $c_{aa}^0 \propto p_a(p_1, p_2)$ where $P_0(|p_1 + p_2|) = \sum_a p_a(p_1, p_2) \mathcal{L}_a(\hat{p}_1 \cdot \hat{p}_2)$. This expansion is the one new ingredient of the three-point case: it is finite in practice (the function is smooth in the opening angle; a few tens of terms resolve the BAO), and it is the only place where the two legs are coupled before the window is involved.

Momenta of the primed estimator. On leg 1 the factors of $\hat{p}_1$ are $Y_{\ell'_1 m'_1}(\hat{p}_1)$ from $\tilde{\mathcal{S}}_{\ell'}$, $Y_{L_1 M_1}(\hat{p}_1)$ from $\mathcal{L}_{L_1}(\hat{p}_1 \cdot \hat{y}_1) = (4\pi/(2L_1 + 1)) \sum_{M_1} Y_{L_1 M_1}(\hat{p}_1) \bar{Y}_{L_1 M_1}(\hat{y}_1)$, $Y_{a\alpha}(\hat{p}_1)$ from Equation 20, and the plane wave $e^{ip_1 \cdot r_1}$. Collapsing the three harmonics with Equation 1 and applying Equation 5,

$$\left\langle e^{ip_1 \cdot r_1} Y_{\ell'_1 m'_1} Y_{L_1 M_1} Y_{a\alpha}(\hat{p}_1) g(p_1) \right\rangle_{i'} = \sum_{\Lambda\mu} G_{\ell'_1 L_1 a \Lambda}^{m'_1 M_1 \alpha \mu} i^\Lambda \bar{g}_\Lambda^{(i')}(r_1) \bar{Y}_{\Lambda\mu}(\hat{r}_1), \quad (21)$$

and the same on leg 2 with $(\ell'_2, m'_2, L_2, M_2, b, \beta, \Lambda', \mu', r_2)$. Because the vertex coefficient $c_{ab}^{L_3}(p_1, p_2)$ depends on both magnitudes, the two radial shell averages do not factorise; they form one two-dimensional kernel,

$$\mathcal{P}_{L_1 L_2 L_3; ab; \Lambda \Lambda'}^{(i' j')}(r_1, r_2) \equiv \frac{\int_{i'} p_1^2 dp_1 \int_{j'} p_2^2 dp_2 j_\Lambda(p_1 r_1) j_{\Lambda'}(p_2 r_2) P_{L_1}(p_1) P_{L_2}(p_2) c_{ab}^{L_3}(p_1, p_2)}{\int_{i'} p_1^2 dp_1 \int_{j'} p_2^2 dp_2}, \quad (22)$$

the analogue of $P_{L\lambda}^{(i)}(s)$, now with the third power spectrum folded in through $c_{ab}^{L_3}$.

Harmonics at the five directions. After these steps the integrand of Equation 16 is a product of harmonics of $\hat{r}_1, \hat{r}_2, \hat{y}_1, \hat{y}_2, \hat{x}$:

$$\hat{r}_1 : \bar{Y}_{\ell'_1 m'_1} \bar{Y}_{\Lambda\mu} = \sum_{\lambda_1 \rho_1} G_{\ell'_1 \Lambda \lambda_1}^{m'_1 \mu \rho_1} Y_{\lambda_1 \rho_1}(\hat{r}_1), \quad \hat{r}_2 : \bar{Y}_{\ell'_2 m'_2} \bar{Y}_{\Lambda'\mu'} = \sum_{\lambda_2 \rho_2} G_{\ell'_2 \Lambda' \lambda_2}^{m'_2 \mu' \rho_2} Y_{\lambda_2 \rho_2}(\hat{r}_2), \quad (23)$$

$$\hat{y}_1 : \bar{Y}_{L_1 M_1}(\hat{y}_1), \quad \hat{y}_2 : \bar{Y}_{L_2 M_2}(\hat{y}_2), \quad \hat{x} : \bar{Y}_{LM} Y_{L'M'} \bar{Y}_{L_3 \gamma} = (-1)^{M'} \sum_{\Lambda_3 \mu_3} G_{LL'L_3 \Lambda_3}^{M, -M', \gamma, \mu_3} Y_{\Lambda_3 \mu_3}(\hat{x}). \quad (24)$$

The phase $(-i)^{\ell_1 + \ell_2} i^{\Lambda + \Lambda'}$ is real: $\Lambda + \Lambda' \equiv a + b \equiv 0 \pmod{2}$.

4.3 Wigner–Eckart and the invariant window function

The coefficient of $Y_{\lambda_1 \rho_1}(\hat{r}_1) Y_{\lambda_2 \rho_2}(\hat{r}_2) \bar{Y}_{L_1 M_1}(\hat{y}_1) \bar{Y}_{L_2 M_2}(\hat{y}_2) Y_{\Lambda_3 \mu_3}(\hat{x})$ is the tensor

$$\begin{aligned} T^{\rho_1 \rho_2 M_1 M_2 \mu_3} = & \sum_{m_1 m_2 M m'_1 m'_2 M' \alpha \beta \gamma \mu \mu'} \begin{pmatrix} \ell_1 & \ell_2 & L \\ m_1 & m_2 & M \end{pmatrix} \begin{pmatrix} \ell'_1 & \ell'_2 & L' \\ m'_1 & m'_2 & M' \end{pmatrix} \langle a\alpha b\beta \mid L_3 \gamma \rangle \\ & \times G_{\ell'_1 L_1 a \Lambda}^{m'_1 M_1 \alpha \mu} G_{\ell'_2 L_2 b \Lambda'}^{m'_2 M_2 \beta \mu'} G_{\ell_1 \Lambda \lambda_1}^{m_1 \mu \rho_1} G_{\ell_2 \Lambda' \lambda_2}^{m_2 \mu' \rho_2} (-1)^{M'} G_{LL'L_3 \Lambda_3}^{M, -M', \gamma, \mu_3}, \end{aligned} \quad (25)$$

built from Gaunt coefficients only, hence rotationally invariant. With five free magnetic indices the space of invariants is no longer one-dimensional: a coupling scheme is needed. The natural one follows the geometry — leg 1 ($\hat{r}_1, \hat{y}_1$), leg 2 ($\hat{r}_2, \hat{y}_2$), and the anchor $\hat{x}$:

$$E_{K_1 K_2}^{\rho_1 M_1 \rho_2 M_2 \mu_3} \equiv \sum_{\kappa_1 \kappa_2} \langle \lambda_1 \rho_1 L_1 M_1 \mid K_1 \kappa_1 \rangle \langle \lambda_2 \rho_2 L_2 M_2 \mid K_2 \kappa_2 \rangle \langle K_1 \kappa_1 K_2 \kappa_2 \mid \Lambda_3 \mu_3 \rangle, \quad (26)$$

in terms of which $T = \sum_{K_1 K_2} \mathcal{T}_{K_1 K_2} E_{K_1 K_2}$ with $\mathcal{T}_{K_1 K_2} = (\sum T E_{K_1 K_2}) / (\sum E_{K_1 K_2}^2)$ (sums over all five indices; the E for different (K_1, K_2) are orthogonal). The invariant window function is the three-point correlation of the window weighted by the corresponding scalar,

$$\begin{aligned} \mathcal{Q}^{\lambda_1 L_1 K_1; \lambda_2 L_2 K_2; \Lambda_3}(r_1, r_2) &\equiv \int d^3x \int d^2\hat{r}_1 d^2\hat{r}_2 W(x)W(x+r_1)W(x+r_2) \\ &\times \sum_{\rho_1 M_1 \rho_2 M_2 \mu_3} E_{K_1 K_2}^{\rho_1 M_1 \rho_2 M_2 \mu_3} Y_{\lambda_1 \rho_1}(\hat{r}_1) \bar{Y}_{L_1 M_1}(\hat{y}_1) Y_{\lambda_2 \rho_2}(\hat{r}_2) \bar{Y}_{L_2 M_2}(\hat{y}_2) Y_{\Lambda_3 \mu_3}(\hat{x}), \end{aligned} \quad (27)$$

the analogue of $\mathcal{Q}_{\Lambda_1 \Lambda_2 \Lambda}(s)$ with one more separation and one more line of sight. It depends on the geometry only.

4.4 Result for class 1

$$\begin{aligned} C_{\ell\ell'}^{(1)}(ij; i'j') &= \frac{(4\pi)^3 HH' \sqrt{NN'}}{I_3^2} \sum_{L_1 L_2 L_3} \frac{(4\pi)^3}{(2L_1+1)(2L_2+1)(2L_3+1)} \sum_{ab} \sum_{\Lambda\Lambda'} \sum_{\lambda_1 \lambda_2 \Lambda_3} \sum_{K_1 K_2} (-1)^{(\Lambda+\Lambda'-\ell_1-\ell_2)/2} \mathcal{T}_{K_1 K_2} \\ &\times \int r_1^2 dr_1 \int r_2^2 dr_2 \bar{j}_{\ell_1}^{(i)}(r_1) \bar{j}_{\ell_2}^{(j)}(r_2) \mathcal{P}_{L_1 L_2 L_3; ab; \Lambda\Lambda'}^{(i'j')}(r_1, r_2) \mathcal{Q}^{\lambda_1 L_1 K_1; \lambda_2 L_2 K_2; \Lambda_3}(r_1, r_2), \end{aligned} \quad (28)$$

where $\mathcal{T}$ carries all the multipole labels $(\ell_1 \ell_2 L; \ell'_1 \ell'_2 L'; L_1 L_2 L_3; ab; \Lambda\Lambda'; \lambda_1 \lambda_2 \Lambda_3)$ and is a pure number computed once from 3- j symbols. The selection rules are those of the Gaunt coefficients: $\Lambda \in \text{tri}(\ell'_1, L_1, a)$, $\lambda_1 \in \text{tri}(\ell_1, \Lambda)$, $K_1 \in \text{tri}(\lambda_1, L_1)$, $\Lambda_3 \in \text{tri}(L, L', L_3) \cap \text{tri}(K_1, K_2)$, and the same for leg 2. Every sum is finite except those over (a, b) , which are the Legendre orders of the third power spectrum in the opening angle.

Shot noise. Each of the three correlators is linear in its window-times-spectrum, so restoring $WP_L \rightarrow WP_L + S\delta_{L0}^K$ at each of the three points gives eight terms: at a leg, S at that point with $L_a = 0$ and $P_{L_a} \rightarrow 1$ inside Equation 22; at the vertex, $S(x)$ with $L_3 = 0$ and $c_{ab}^0 \rightarrow \delta_{a0}\delta_{b0}\sqrt{4\pi}$ (the constant $Y_{00}P_0 \rightarrow Y_{00}$). The window functions become $\mathcal{Q}^{XYZ}$ with $X, Y, Z \in \{W, S\}$ for the anchor and the two legs. The estimator's own shot-noise subtraction removes the self-pairs inside each estimator and leaves this structure intact, as it does for the power spectrum (Sugiyama et al. 2020, sec. 2.3).

5 Class 2: a line of sight pairs with a shell galaxy

For the pairing $(11')(23')(32')$ the correlators are Equation 12 for (q_1, p_1) at y_1 , Equation 13 for (q_2, p_3) at y_2 with $p_3 = p_1 + p_2$, and Equation 12 read backwards for $(x, -p_2)$: $\int d^3y'_2 e^{ip_2 \cdot y'_2} \langle F(x)F(y'_2) \rangle = e^{ip_2 \cdot x} W(x)P(p_2; \hat{x})$. The line-of-sight harmonic of the primed estimator now sits at $\hat{y}_2$. The phases combine into

$$e^{-i(q_1-p_1) \cdot r_1} e^{-i(q_2+p_1+p_2) \cdot r_2} = e^{-iq_1 \cdot r_1} e^{-iq_2 \cdot r_2} e^{-ip_2 \cdot r_2} e^{-ip_1 \cdot (r_2-r_1)}, \quad (29)$$

and the master formula is

$$\begin{aligned} C_{\ell\ell'}^{(2)} &= \frac{(4\pi)^3 HH' \sqrt{NN'}}{I_3^2} \langle \langle \mathcal{I}^{(2)} \rangle \rangle_{ij i' j'}, \quad \mathcal{I}^{(2)} \equiv \int d^3x d^3r_1 d^3r_2 W(x)W(x+r_1)W(x+r_2) \\ &\times e^{-i(q_1-p_1) \cdot r_1} e^{-i(q_2+p_1+p_2) \cdot r_2} \bar{\mathcal{S}}_{\ell}(\hat{q}_1, \hat{q}_2, \hat{x}) \bar{\mathcal{S}}_{\ell'}(\hat{p}_1, \hat{p}_2, \hat{y}_2) P(p_1; \hat{y}_1) P(p_1+p_2; \hat{y}_2) P(p_2; \hat{x}). \end{aligned} \quad (30)$$

The other three class-2 pairings follow by the two relabellings $(p_1, \ell'_1, i') \leftrightarrow (p_2, \ell'_2, j')$ and $(q_1, \ell_1, i, r_1) \leftrightarrow (q_2, \ell_2, j, r_2)$.

Box limit. $\int d^3r_1 d^3r_2$ gives $(2\pi)^6 \delta_D(q_1 - p_1) \delta_D(q_2 + p_1 + p_2)$: the shell of q_2 must contain the third side of the primed triangle. With the shell averages,

$$\begin{aligned} &\langle \langle \mathbb{1}[|p_1 + p_2| \in j] \bar{\mathcal{S}}_{\ell}(\hat{p}_1, -\hat{p}_3, \hat{n}) \bar{\mathcal{S}}_{\ell'}(\hat{p}_1, \hat{p}_2, \hat{n}) \\ &\times P(p_1)P(p_2)P(|p_1 + p_2|) \rangle \rangle_{ij i' j'}, \end{aligned} \quad C_{\text{box}}^{(2)} = \frac{(4\pi)^3 HH' \sqrt{NN'}}{VV_i V_j} \delta_{ii'} \quad (31)$$

the terms of eq. (49) of Sugiyama et al. (2020) in which k_3 falls in a bin. On the diagonal $j' = j$ the indicator selects the configurations with $|p_1 + p_2| \in j$, a fraction $\sim \Delta k/(2k)$ of the shell of p_2 (the

third side ranges over $|p_1 - p_2| \leq p_3 \leq p_1 + p_2$, a width $\sim 2k$: class 2 is suppressed by $\sim \Delta k/k$ on the diagonal, 5% at $k = 0.1$ for $\Delta k = 0.005$, and it is the only Gaussian contribution off the diagonal, where it is of the same relative size.

Structure. In the anchored basis the two plane waves of p_1 factorise, $e^{-ip_1 \cdot (r_2 - r_1)} = e^{ip_1 \cdot r_1} e^{-ip_1 \cdot r_2}$, so the $\hat{p}_1$ average couples harmonics of $\hat{r}_1$ and $\hat{r}_2$:

$$\begin{aligned} \int \frac{d\Omega_{p_1}}{4\pi} e^{ip_1 \cdot r_1} e^{-ip_1 \cdot r_2} [Y \dots (\hat{p}_1)] &= \sum_{\lambda\lambda'} i^{\lambda-\lambda'} j_\lambda(p_1 r_1) j_{\lambda'}(p_1 r_2) \\ &\times \int \frac{d\Omega_{p_1}}{4\pi} Y_\lambda(\hat{p}_1) \bar{Y}_{\lambda'}(\hat{p}_1) [Y \dots (\hat{p}_1)] \bar{Y}_\lambda(\hat{r}_1) Y_{\lambda'}(\hat{r}_2). \end{aligned} \quad (32)$$

The Gaunt coefficient bounds $|\lambda - \lambda'|$ by the rank of the other harmonics of $\hat{p}_1$, but not λ itself: the expansion in harmonics of the two anchored separations does not terminate (it is the Bessel addition theorem for $j_\lambda(p_1 |r_2 - r_1|)$, which converges once $\lambda > p_1 \max(r_1, r_2)$). The rest of the derivation is as in class 1, with $\mathcal{Q}$ functions whose line-of-sight labels sit at $\hat{x}$ for L and at $\hat{y}_2$ for L' . Which orders are needed in practice is decided by the window, not by the kernel: see the next section.

6 Computing the window function from pair sums

The scalar in Equation 27 couples harmonics of $(\hat{r}_1, \hat{y}_1)$ into K_1 , of $(\hat{r}_2, \hat{y}_2)$ into K_2 , and the two into Λ_3 at the anchor. Therefore

$$\mathcal{Q}^{\dots}(r_1, r_2) = \sum_{\kappa_1 \kappa_2 \mu_3} \langle K_1 \kappa_1 K_2 \kappa_2 \mid \Lambda_3 \mu_3 \rangle \int d^3x W(x) Y_{\Lambda_3 \mu_3}(\hat{x}) A_{K_1 \kappa_1}^{(1)}(x; r_1) A_{K_2 \kappa_2}^{(2)}(x; r_2), \quad (33)$$

$$A_{K\kappa}^{(a)}(x; r) \equiv \int d^2\hat{r} W(x+r) \sum_{\rho M} \langle \lambda_a \rho L_a M \mid K\kappa \rangle Y_{\lambda_a \rho}(\hat{r}) \bar{Y}_{L_a M}(\hat{y}), \quad (34)$$

and $A_{K\kappa}^{(a)}(x; r)$ is a **pair** quantity: the harmonic moments of the window on the shell of radius r around the anchor x , accumulated from the pairs $(x, x+r)$ exactly as the pair counts of `WindowLibrary`, in bins of r . The three-point function is then the sum over anchors of the product of two such moments. This is the Slepian–Eisenstein construction of the three-point function, and it needs no triplets: the cost is (anchors) $\times$ (neighbours per anchor) for the moments, plus (anchors) $\times$ (bins of r_1) $\times$ (bins of r_2) $\times$ (label pairs) for the products. The same construction holds in class 2, where the two plane waves of p_1 are each attached to one anchored side.

What bounds the ranks. The exact expansions contain unbounded sums: over (a, b) in class 1 through $\lambda_{1,2} \in \text{tri}(\ell, \Lambda)$, and over (λ, λ') in class 2. The product of a kernel of rank λ in $\hat{r}_1$ with $\mathcal{Q}$ integrates to zero unless $\mathcal{Q}$ has that rank, and $\mathcal{Q}$ is smooth in the opening angle of the triangle: the window varies on hundreds of Mpc/h , so $\mathcal{Q}^{\dots}(r_1, r_2)$ at ranks beyond $\lambda \sim 6-8$ is negligible for r_1, r_2 below the window scale, and exactly zero in a periodic box, where only the fully isotropic component $\lambda_1 = \lambda_2 = \Lambda_3 = L_a = 0$ survives and reproduces the box formulae above (the isotropic component of the kernel is the full angular average, which is what the delta functions select). Truncating the window's ranks at λ_w therefore truncates every sum, with an error set by the window's angular smoothness and not by the kernels. The (a, b) sums of class 1 are then bounded by $\lambda_w + \ell + L_a$, and the class-2 (λ, λ') sums by λ_w ; alternatively the kernel's projection on the low-rank basis is computed by direct angular quadrature of Equation 16 and Equation 30 at each (r_1, r_2) , which avoids the Gaunt algebra for the primed momenta altogether. The convergence in λ_w is checked by computing $\mathcal{Q}$ to a higher rank, at pair-count cost.

Cost. With $n_{\text{sub}} = 5000$ far randoms as anchors and neighbours: 2.5×10^7 pair evaluations for the moments (as now), and for the products $5000 \times n_r^2 \times (\text{label pairs}) \times (\text{magnetic contractions})$. For $\lambda_w = 4$, $L_a \leq 2$ and the diagonal elements the label pairs are ~ 2500 and the contraction $\sim 10^2$, i.e. $\sim 10^{11}$ operations per 100×100 radial grid: minutes on a GPU, an hour on a node. The near-pair regime ($s <$

s_{split}) uses the tree neighbours as now. Rotating each anchor's frame to $\hat{\mathbf{x}} = \hat{\mathbf{z}}$ makes $\mu_3 = 0$ and removes one magnetic sum.

7 B_{000} and B_{202} on the diagonal

B_{000} . $\ell_1 = \ell_2 = L = 0$ on both sides, $H = 1$, $N = 1$. The six Y_{00} factors supply $(4\pi)^{-3}$ and cancel the $(4\pi)^3$ of the prefactor. The $\hat{r}_1$ collapse forces $\lambda_1 = \Lambda$, the $\hat{r}_2$ one $\lambda_2 = \Lambda'$, and the anchor $\Lambda_3 = L_3$, so Equation 28 becomes

$$C_{000}^{(1)}(ij; i'j') = \frac{1}{I_3^2} \sum_{L_1 L_2 L_3} \frac{(4\pi)^3}{(2L_1 + 1)(2L_2 + 1)(2L_3 + 1)} \sum_{ab\Lambda\Lambda'K_1K_2} (-1)^{(\Lambda+\Lambda')/2} \mathcal{T}_{K_1K_2} \times \int \int r_1^2 r_2^2 dr_1 dr_2 \bar{j}_0^{(i)}(r_1) \bar{j}_0^{(j)}(r_2) \mathcal{P}_{L_1 L_2 L_3; ab; \Lambda\Lambda'}^{(i'j')} \mathcal{Q}^{\Lambda L_1 K_1; \Lambda' L_2 K_2; L_3}, \quad (35)$$

with $\Lambda \in \text{tri}(L_1, a)$, $\Lambda' \in \text{tri}(L_2, b)$. In real space ($L_1 = L_2 = L_3 = 0$, $a = b$, $\Lambda = \Lambda' = K_1 = K_2 = a$) the scalar of Equation 27 is proportional to $\mathcal{L}_a(\hat{r}_1 \cdot \hat{r}_2)$: the window enters through its three-point function projected on Legendre polynomials of the opening angle, and the sum over a is the Legendre expansion of $P(|p_1 + p_2|)$; in the box only $a = 0$ survives and the kernel reduces to the angle-averaged $P(|q_1 + q_2|)$. In redshift space the ranks are $L_{1,2,3} \in \{0, 2, 4\}$, with $\Lambda_3 = L_3$: the window's line-of-sight dependence at the anchor is the quadrupole and hexadecapole of the vertex correlator, and at the legs those of the two leg correlators.

B_{202} . $\ell_1 = 2$, $\ell_2 = 0$, $L = 2$ (and the same primed), $\mathcal{S}_{202}(\hat{k}_1, \hat{k}_2, \hat{\mathbf{n}}) = \mathcal{L}_2(\hat{k}_1 \cdot \hat{\mathbf{n}})$, $H_{202} = -\sqrt{1/5}$, $N = 25$. The 3- j symbols are $\begin{pmatrix} 2 & 0 & 2 \\ m_1 & 0 & -m_1 \end{pmatrix} = (-1)^{m_1}/\sqrt{5}$ and reduce the double sums to one magnetic index per estimator. The index sets: at $\hat{r}_2$ still $\lambda_2 = \Lambda'$ (because $\ell_2 = 0$); at $\hat{r}_1$, $\lambda_1 \in \text{tri}(2, \Lambda)$, with $\Lambda \in \text{tri}(2, L_1, a)$; at the anchor $\Lambda_3 \in \text{tri}(2, 2, L_3) \cap \text{tri}(K_1, K_2)$, up to $4 + L_3$. The quadrupole weight on leg 1 is a rank-2 harmonic of $\hat{r}_1$ in the unprimed kernel and a rank-2 harmonic of $\hat{p}_1$ in the primed one; the anchor carries the two line-of-sight quadrupoles of the estimators and the vertex anisotropy. For B_{202} the second class-1 pairing is not the mirror of the first ($\ell_1 \neq \ell_2$), so both must be evaluated; the diagonal of B_{202} at $k_1 = k_2$ receives the swapped term with the roles of the two legs exchanged.

Recipe for the diagonal. (i) Pair counts with the anchored moments $A_{K\kappa}^{(a)}(x; r)$ for $\lambda \leq \lambda_w$, $L_a \in \{0, 2, 4\}$, and the eight (W, S) combinations; (ii) the invariants $\mathcal{Q}$ for the labels above, contracted per anchor; (iii) the kernels: $\bar{j}_\ell^{(i)}$ from the shells, the two-dimensional $\mathcal{P}$ of Equation 22 from the model multipoles and the bipolar coefficients $c_{ab}^{L_3}(p_1, p_2)$ (a one-dimensional integral per (a, b, L_3) and per node of the two shells), and for class 2 the projected kernel by quadrature; (iv) the coefficients $\mathcal{T}$ from 3- j symbols; (v) the double radial integral. The class-1 terms give the diagonal to $O(\Delta k/k)$; adding the four class-2 terms completes the Gaussian part, on and off the diagonal.

8 Non-Gaussian terms: PB , BB and PT

The connected correlators localise. In the local approximation a connected three-point function of three fields puts all three within a correlation length of one point, so with the phases of the fields written out,

$$\langle F(k_a) F(k_b) F(k_c) \rangle_c \approx \int d^3 y e^{-i(k_a + k_b + k_c) \cdot y} \omega^3(y) B(k_a, k_b; \hat{\mathbf{y}}), \quad (36)$$

where $B(k_a, k_b; \hat{\mathbf{n}})$ is the bispectrum with third momentum $-(k_a + k_b)$ and line of sight $\hat{\mathbf{n}}$. A configuration-space field $F(x)$ that carries an estimator phase $e^{iK \cdot x}$ is a Fourier field of momentum $-K$ at the point x ; if it is the anchor the y integral is absent and $\omega^3(x)$ appears instead, exactly as $W(x)$ in Equation 14. The same rule gives the connected four-point function with $\omega^4(y)$ and the trispectrum $T(k_a, k_b, k_c; \hat{\mathbf{y}})$, the fourth momentum fixed by conservation. In the Sugiyama basis $B(k_a, k_b; \hat{\mathbf{n}}) = \sum_{\ell''} B_{\ell''}(k_a, k_b) \mathcal{S}_{\ell''}(\hat{k}_a, \hat{k}_b, \hat{\mathbf{n}})$ with $\ell'' = (\ell_1'' \ell_2'' L'')$.

Two consequences hold for every term below. First, each term has exactly one plane wave and one separation, between the point of one connected factor and the point of the other, so **the window enters through a two-point function**, a pair count with the weights (ω^2, ω^3) , (ω^3, ω^3) or (ω^2, ω^4) in place of (ω^2, ω^2) : the functions $\mathcal{Q}_{\Lambda_1 \Lambda_2 \Lambda}(s)$ of the power-spectrum note with other weights, and no three-point window function. Second, with one separation and two line-of-sight points the invariant space is one-dimensional and the Wigner–Eckart step is the 3- j symbol of the power-spectrum note. The difficulty moves to the kernels: the bispectrum and trispectrum models, whose angular dependence in the shells is reduced by the same bipolar expansion as Equation 20.

We write the power-spectrum estimator with the line of sight at the configuration-space point, $I_2 \equiv \int d^3x \omega^2$, and shells i of k :

$$\hat{P}_\ell(i) = \frac{2\ell+1}{I_2} \left\langle \left\langle \int d^3x e^{-ik \cdot x} \mathcal{L}_\ell(\hat{k} \cdot \hat{x}) F(x) F(-k) \right\rangle \right\rangle_i, \quad (37)$$

so that $F(x)$ carries momentum k at x and $F(-k)$ is the Fourier field; its fields are labelled $a \equiv F(-k)$ and $b \equiv F(x)$.

8.1 The PB cross-covariance

$\langle \hat{P}_\ell(i) \hat{B}'_{\ell'}(i', j') \rangle_c$ has five fields, a, b and $1', 2', 3'$. There is no Gaussian term (five fields), and the leading term pairs one field of $\hat{P}$ with one of $\hat{B}'$ and leaves the other three connected; the pairings with the pair inside $\hat{B}'$ are the beat-mode terms (window at $p_1 + p_2$, as in the Gaussian case) and the pairing (ab) is disconnected. The six cross pairings come in four structures (the mirror $1' \leftrightarrow 2'$ doubles the first two):

$$\begin{aligned} (a1')(b2'3'), \quad (b1')(a2'3') \quad \text{leg: } k \approx p_1 \text{ (and } p_2), \\ (a3')(b1'2'), \quad (b3')(a1'2') \quad \text{third side: } k \approx |p_1 + p_2|. \end{aligned} \quad (38)$$

For $(a1')(b2'3')$ the two-point factor is Equation 12 with $q = -k$: $\int d^3y e^{i(k+p_1) \cdot y} W(y) P(p_1; \hat{y})$, and the connected factor, anchored at the line-of-sight point x of $\hat{P}$ with the phases $e^{-ik \cdot x}$ of $\hat{P}$ and $e^{-i(p_1+p_2) \cdot x'}$ of $\hat{B}'$, is by Equation 36

$$\int d^3x' e^{-i(p_1+p_2) \cdot x'} \langle F(x) F(-p_2) F(x') \rangle_c \approx \omega^3(x) e^{-ip_1 \cdot x} B(-p_2, p_1 + p_2; \hat{x}), \quad (39)$$

with the line-of-sight harmonic of $\hat{B}'$ moved from $\hat{x}'$ to $\hat{x}$. The phases combine into $e^{i(k+p_1) \cdot s}$, $s \equiv y - x$, and

$$\begin{aligned} C_{\ell, \ell'}^{PB, (a1')}(i; i' j') = \frac{(2\ell+1)(4\pi)^{3/2} H' \sqrt{N'}}{I_2 I_3} \langle \langle \mathcal{I}^{PB} \rangle \rangle_{ii' j'}, \quad \mathcal{I}^{PB} \equiv \int d^3x d^3s \omega^3(x) W(x+s) e^{i(k+p_1) \cdot s} \\ \times \mathcal{L}_\ell(\hat{k} \cdot \hat{x}) \tilde{\mathcal{S}}_{\ell'}(\hat{p}_1, \hat{p}_2, \hat{x}) P(p_1; \hat{y}) B(-p_2, p_1 + p_2; \hat{x}), \end{aligned} \quad (40)$$

$\hat{y} = (x+s)/|x+s|$. The box limit, $\int d^3s \rightarrow (2\pi)^3 \delta_D(k+p_1)$, gives $\delta_{ii'}/V_i$ and $P(k)B(k, -p_2, p_2 - k)$ at $\hat{k} = -\hat{p}_1$: the PB terms of Sugiyama et al. (2020) with the leg coincidence; $(b1')(a2'3')$ gives the same box limit with W at the anchor and ω^3 at the far point, $P(p_1; \hat{x})B(-k, -p_2; \hat{y})$, the analogue of the two orderings $T_{AC, DB}$ and $T_{AD, CB}$ of the power-spectrum note. The third-side pairings carry $P(p_1 + p_2; \cdot)$ in the two-point factor and $B(-p_1, -p_2; \cdot)$ in the connected one, with the plane wave $e^{i(k-p_1-p_2) \cdot s}$ (or $k + p_1 + p_2$): in the box they need $|p_1 + p_2| \in i$, a fraction $\sim \Delta k/(2k)$ of the p_2 shell on the “diagonal” $i = j'$.

Harmonic reduction of Equation 40. $\mathcal{L}_\ell(\hat{k} \cdot \hat{x}) = (4\pi/(2\ell+1)) \sum_m Y_{\ell m}(\hat{k}) \bar{Y}_{\ell m}(\hat{x})$ and the conjugate of Equation 5 give $\langle e^{ik \cdot s} Y_{\ell m}(\hat{k}) \rangle_i = i^\ell \bar{j}_\ell^{(i)}(s) Y_{\ell m}(\hat{s})$. The bispectrum model is expanded in its own basis, $B(-p_2, p_1 + p_2; \hat{x}) = \sum_{\ell''} B_{\ell''}(p_2, |p_1 + p_2|) \frac{(4\pi)^{3/2}}{H'' \sqrt{N''}} \tilde{\mathcal{S}}_{\ell''}(-\hat{p}_2, \hat{p}_3, \hat{x})$, and the factor that depends on $\hat{p}_3$ is bipolar in $(\hat{p}_1, \hat{p}_2)$ exactly as in Equation 20,

$$Y_{\ell_2'' m_2''}(\hat{p}_3) B_{\ell''}(p_2, |p_1 + p_2|) = \sum_{ab} c_{ab}^{\ell''}(p_1, p_2) [Y_a(\hat{p}_1) \otimes Y_b(\hat{p}_2)]_{\ell_2'' m_2''}. \quad (41)$$

Then $\hat{p}_2$ carries $Y_{\ell_2'}$, $Y_{\ell_1''}$ and Y_b and no plane wave: its shell average is a Gaunt coefficient, $G_{\ell_2' \ell_1'' b}^{m_2' m_1'' \beta} / (4\pi)$, with $b \in \text{tri}(\ell_2', \ell_1'')$. $\hat{p}_1$ carries $Y_{\ell_1'}$, Y_{L_1} from $P(p_1; \hat{y})$, Y_a and the plane wave $e^{ip_1 \cdot s}$: as on leg 1 of class 1, $\sum_{\Lambda} G_{\ell_1' L_1 a \Lambda}^{m_1' M_1 \alpha \mu} i^{\Lambda} \bar{g}_{\Lambda}^{(i')}(s) \bar{Y}_{\Lambda \mu}(\hat{s})$. Three directions remain, $\hat{s}$ with $Y_{\ell} \bar{Y}_{\Lambda} \rightarrow \lambda$, $\hat{x}$ with the three line-of-sight harmonics $\bar{Y}_{\ell}(\hat{x}) Y_{L' M'}(\hat{x}) Y_{L'' M''}(\hat{x}) \rightarrow \Lambda_x$, and $\hat{y}$ with $\bar{Y}_{L_1 M_1}$. The coefficient tensor has three magnetic indices and is $\mathcal{T}_{\lambda \Lambda_x L_1} \begin{pmatrix} \lambda & \Lambda_x & L_1 \\ \rho & \mu_x & M_1 \end{pmatrix}$ after the conjugated harmonics are written as $\bar{Y}_{LM} = (-1)^M Y_{L, -M}$, so

$$C_{\ell, \ell'}^{PB, (a1')}(i; i' j') = \frac{(2\ell + 1)(4\pi)^{3/2} H' \sqrt{N'}}{I_2 I_3} \sum_{\ell'' L_1 ab \Lambda} \frac{(4\pi)^{5/2}}{(2\ell + 1)(2L_1 + 1) H'' \sqrt{N''}} \sum_{\lambda \Lambda_x} i^{\ell + \Lambda} \mathcal{T}_{\lambda \Lambda_x L_1} \times \int s^2 ds \bar{j}_{\ell}^{(i)}(s) \mathcal{P}_{L_1 \ell''; ab; \Lambda}^{(i' j')}(s) \mathcal{Q}_{\Lambda_x L_1 \lambda}^{\omega^3, W}(s), \quad (42)$$

$$\mathcal{P}_{L_1 \ell''; ab; \Lambda}^{(i' j')}(s) \equiv \frac{\int_{i'} p_1^2 dp_1 \int_{j'} p_2^2 dp_2 j_{\Lambda}(p_1 s) P_{L_1}(p_1) c_{ab}^{\ell''}(p_1, p_2)}{\int_{i'} p_1^2 dp_1 \int_{j'} p_2^2 dp_2}, \quad (43)$$

with $\mathcal{Q}_{\Lambda_x L_1 \lambda}^{\omega^3, W}(s) = \int d^3 x \int d^2 \hat{s} \omega^3(x) W(x + s) \sum (3-j) Y_{\Lambda_x}(\hat{x}) Y_{L_1}(\hat{y}) Y_{\lambda}(\hat{s})$ the window function of the power-spectrum note with the weights (ω^3, ω^2) , and $\mathcal{T}$ the contraction of the Gaunt coefficients of the chain above (the same product as Equation 25 with one leg, three directions and the model's 3- j symbol in place of the second estimator's). The phase is real: $\Lambda \equiv \ell_1' + L_1 + a$, $a + b \equiv \ell_2''$, $b \equiv \ell_2' + \ell_1'' \pmod{2}$, so $\ell + \Lambda \equiv L' + L'' \equiv 0$. The radial kernel is one-dimensional: there is one separation. The third-side pairings have the same form with $P_{L_1}(p_1) \rightarrow P_{L_1}(|p_1 + p_2|)$ inside the bipolar coefficient and $B_{\ell''}(p_1, p_2)$ with both arguments on the shells, i.e. with c built from $Y(\hat{p}_3) P_{L_1}(p_3)$ and the plane wave of p_3 carried by the pair $(\hat{p}_1, \hat{p}_2)$ jointly, which is where the box factor $\Delta k / (2k)$ comes from.

8.2 The BB term

The six fields split into two connected triples. Of the ten splittings, $(123)(1'2'3')$ is disconnected and the other nine each have one unprimed field with two primed ones. Writing the singleton of the unprimed estimator and the pair of the primed one, the coincidence that the plane wave enforces is

$$\begin{aligned} (3 | 1'2') : k_3 \approx k_3', \quad (3 | 1'3'), (3 | 2'3') : k_3 \approx k_2', k_1', \quad (1 | 1'2'), (2 | 1'2') : k_1, k_2 \approx k_3', \\ (1 | 2'3'), (1 | 1'3'), (2 | 1'3'), (2 | 2'3') : k_1 \approx k_1', k_1 \approx k_2', k_2 \approx k_2', k_2 \approx k_1', \end{aligned} \quad (44)$$

where $k_3 = |q_1 + q_2|$ and $k_3' = |p_1 + p_2|$ are the unbinned third sides. The four leg-leg pairings give the $\delta_{ii'}$ -type terms of eq. (32) of Sugiyama et al. (2020) with full weight; the five that involve a third side are suppressed by $\sim \Delta k / (2k)$ on the diagonal, as class 2 is. For $(1 | 2'3')$, with the first triple anchored at x' and the second at x ,

$$\begin{aligned} \int d^3 x' e^{-i(p_1 + p_2) \cdot x'} \langle F(q_1) F(-p_1) F(x') \rangle_c \approx \omega^3(x') e^{-i(q_1 + p_2) \cdot x'} B(q_1, -p_1; \hat{x}'), \\ e^{i(q_1 + q_2) \cdot x} \langle F(q_2) F(x) F(-p_2) \rangle_c \approx \omega^3(x) e^{i(q_1 + p_2) \cdot x} B(q_2, -p_2; \hat{x}), \end{aligned} \quad (45)$$

so with $s = x' - x$

$$\begin{aligned} C_{\ell \ell'}^{BB, (1 | 2'3')}(ij; i' j') = \frac{(4\pi)^3 H H' \sqrt{N N'}}{I_3^2} \langle \langle \mathcal{I}^{BB} \rangle \rangle_{ij i' j'}, \quad \mathcal{I}^{BB} \equiv \int d^3 x d^3 s \omega^3(x) \omega^3(x + s) e^{-i(q_1 + p_2) \cdot s} \\ \times \bar{\mathcal{S}}_{\ell}(\hat{q}_1, \hat{q}_2, \hat{x}) \tilde{\mathcal{S}}_{\ell'}(\hat{p}_1, \hat{p}_2, \hat{y}) B(q_2, -p_2; \hat{x}) B(q_1, -p_1; \hat{y}), \end{aligned} \quad (46)$$

$\hat{y} = (x + s) / |x + s|$: the coincidence is $q_1 \approx -p_2$ ($i = j'$), each estimator's line of sight sits at its own anchor, the unprimed estimator's line of sight with $B(q_2, -p_2; \hat{x})$ and the primed one's with $B(q_1, -p_1; \hat{y})$. The box limit is $(\delta_{ij'} / V_i) \langle \langle \dots B(q_2, -p_2, p_2 - q_2) B(q_1, -p_1, p_1 - q_1) \rangle \rangle$ with $\hat{p}_2 = -\hat{q}_1$, the first term of eq. (32).

Reduction. Each bispectrum is expanded in its basis and the third-side arguments, here $|p_2 - q_2|$ and $|p_1 - q_1|$ through $B(q_2, -p_2; \hat{\mathbf{x}}) = \sum B_{\ell''}(q_2, p_2) \mathcal{S}_{\ell''}(\hat{q}_2, -\hat{p}_2, \hat{\mathbf{x}})$ with the magnitudes of the two binned momenta as arguments, need no bipolar expansion when both arguments are shell momenta: $B_{\ell''}(q_2, p_2)$ depends on the magnitudes only and the opening angle is in $\mathcal{S}_{\ell''}$. In the pairings that involve a third side, one bispectrum argument is $|q_1 + q_2|$ or $|p_1 + p_2|$ and the bipolar expansion of Equation 20 applies to $Y(\hat{q}_3) B_{\ell''}(\cdot, q_3)$. In Equation 46 the momenta without a plane wave, $\hat{q}_2$ and $\hat{p}_1$, carry two harmonics each ($\hat{\mathcal{S}}$ and the model) and average to Gaunt coefficients; $\hat{q}_1$ and $\hat{p}_2$ carry the plane wave $e^{-i(q_1+p_2)\cdot s}$ and give $\bar{j}_{\ell_1+\dots}^{(i)}$ and $\bar{j}_{\ell_1+\dots}^{(j')}$ through Equation 5, two Bessel averages at the same s and no model power spectrum on the plane wave. The three remaining directions are $\hat{s}$, $\hat{\mathbf{x}}$ (with $\bar{Y}_L$, $Y_{L''}$) and $\hat{y}$ (with $Y_{L'}$, $Y_{L''}$ of the two models' lines of sight), so

$$C^{BB} = \sum(\dots) \mathcal{T}_{\Lambda\Lambda_x\Lambda_y} \int s^2 ds \mathcal{K}^{(ijj')}(s) \mathcal{Q}_{\Lambda_x\Lambda_y\Lambda}^{\omega^3, \omega^3}(s), \quad (47)$$

with a kernel $\mathcal{K}(s)$ built from products of $B_{\ell''}(q, p)$ averaged over the two shells of each bispectrum and Bessel functions of the two plane-wave momenta, and $\mathcal{Q}^{\omega^3, \omega^3}$ the pair-count window function with weights ω^3 at both points. On the diagonal $(i, j) = (i', j')$ the four leg-leg pairings contribute; each is a product of two bispectra at the binned sides, $B(k_1, k_2, k_3)^2$ in the box, so this is the term that grows fastest with the signal and the reason the Gaussian-only covariance fails at $k \gtrsim 0.1$.

8.3 The PT term

The six fields split into a cross pair and a connected four-point function of the other four (the pairs inside one estimator are again beat-mode terms). The nine cross pairs have the same coincidence table as the Gaussian pairings: four leg-leg, $(11')$, $(12')$, $(21')$, $(22')$; two leg-third, $(13')$, $(23')$; two third-leg, $(31')$, $(32')$; one third-third, $(33')$. For $(11')$ the pair is Equation 12, $\int d^3y e^{-i(q_1-p_1)\cdot y} W(y) P(p_1; \hat{y})$, and the four-point function, anchored at x with both configuration-space fields,

$$\begin{aligned} & \int d^3x' e^{i(q_1+q_2)\cdot x - i(p_1+p_2)\cdot x'} \langle F(q_2) F(x) F(-p_2) F(x') \rangle_c \\ & \approx \omega^4(x) e^{i(q_1-p_1)\cdot x} T(q_2, -p_2, p_1 + p_2; \hat{\mathbf{x}}), \end{aligned} \quad (48)$$

the trispectrum with fourth momentum $-(q_2 + p_1)$, which closes the quadrilateral exactly when $q_1 = p_1$. With $s = y - x$,

$$\begin{aligned} C_{\ell\ell'}^{PT, (11')}(ij; i'j') &= \frac{(4\pi)^3 H H' \sqrt{N N'}}{I_3^2} \langle \langle \mathcal{I}^{PT} \rangle \rangle_{ijj'j'}, \quad \mathcal{I}^{PT} \equiv \int d^3x d^3s \omega^4(x) W(x+s) e^{-i(q_1-p_1)\cdot s} \\ & \times \bar{\mathcal{S}}_{\ell}(\hat{q}_1, \hat{q}_2, \hat{\mathbf{x}}) \tilde{\mathcal{S}}_{\ell'}(\hat{p}_1, \hat{p}_2, \hat{\mathbf{x}}) P(p_1; \hat{y}) T(q_2, -p_2, p_1 + p_2; \hat{\mathbf{x}}), \end{aligned} \quad (49)$$

whose box limit $(\delta_{ii'}/V_i) \langle \langle \dots P(q_1) T(q_2, -p_2, q_1 + p_2, -q_1 - q_2) \rangle \rangle$ is the PT term of eq. (33) of Sugiyama et al. (2020). The window enters through $\mathcal{Q}^{\omega^4, W}(s)$, the pair count with weights (ω^4, ω^2) , and the plane wave carries only the beat momentum $q_1 - p_1$: the whole angular structure of T — it depends on $\hat{q}_2$, $\hat{p}_2$, $\hat{p}_1$ and $\hat{\mathbf{x}}$ jointly — is averaged over the shells of q_2 and p_2 with the estimator harmonics, and only its dependence on $\hat{p}_1$ (through the plane wave, ranks bounded by the window) and on $\hat{\mathbf{x}}$ (the line-of-sight ranks) survives into the Wigner–Eckart contraction. In practice this is a Monte Carlo over the shell triples, as in `thecov.trispectrum`, with the integrand projected on $Y_{\Lambda}(\hat{p}_1)$ and on the line-of-sight harmonics at $\hat{\mathbf{x}}$ before the s integral; the shell of q_1 enters only through $\bar{j}_{\ell_1}^{(i)}(s)$.

The squeezed limit. The beat momentum $\varepsilon = q_1 - p_1$ is not zero under the window: T is evaluated on quadrilaterals whose momenta sum to $-\varepsilon$, and for $|\varepsilon| \lesssim k_w$ it contains the response terms $\propto P(\varepsilon)$ of the connected four-point function. Integrated against $|\tilde{W}(\varepsilon)|^2$ these are the super-sample covariance of the bispectrum, the analogue of the T_0 piece of the power-spectrum covariance. Either T is evaluated at $\varepsilon = 0$ (the standard PT term) and the super-sample term is added separately, with its own responses, or the full ε dependence is kept and no separate super-sample term is added. Doing both double counts; `thecov` does the former for P .

8.4 Shot noise, models and calibration

Shot noise. Each connected correlator carries its Poisson terms, which in the local approximation change the weights: for the three-point function $\omega^3 B \rightarrow \omega^3 B + S \omega [P(k_a) + P(k_b) + P(k_c)] + (1 - \alpha^2) \bar{n} w^3$, with $S = (1 + \alpha) \bar{n} w^2$ for the one-shared-galaxy terms (the pair window of the two coincident fields) and the three-shared-galaxy constant; for the four-point function the terms with one, two and three shared galaxies follow the same rule ($\omega^4 T + S \omega^2 B + \dots$). `thecov.discreteness` has the one- and two-shared-galaxy pieces. The pair counts needed are therefore those with weights $\omega^2, \omega^3, \omega^4$ and $S\omega, S\omega^2, S^2$, which `WindowLibrary` can accumulate in one pass.

Models. Tree level throughout, with the kernels already in `thecov`: $B = 2Z_1 Z_1 Z_2 PP + 2$ perms. (SCF99, b_1, b_2, b_{s^2}, f , in `discreteness.py` and `ssc.py`) and $T = T_{2211} + T_{3111}$ (`trispectrum.py`, Galileon bias), on the same damped P_{lin} as the Gaussian term, with the finger-of-god damping applied to every Z_1 leg so that P, B and T are damped consistently. The BB and PB kernels need $B_{\ell''}(k, k')$ in the Sugiyama basis on a two-magnitude grid, obtained once by angular quadrature of the model; the PT kernel needs T on general quadrilaterals, which the kernel code gives but the current integration driver (parallelograms $(k, -k, k', -k')$) does not. Loop corrections and the connected five- and six-point terms (P_5 in PB , P_6 in BB) are not worth their cost here: they are corrections to terms that are themselves corrections.

Calibration from the data. $B_{\ell''}(k, k')$ can be taken from the measured multipoles, binned on the covariance grid and lightly smoothed, with the shot-noise pieces left in, as the measured P replaces the model in the Gaussian term; the window convolution of the measurement matters only within a few k_w of the window scale, and in noisy bins the tree-level model with b_1, b_2, f fitted to $\hat{B}$ is safer. T cannot be measured, but the configurations the PT term integrates over can: the squeezed part is the correlation of the local power spectrum with the local mean density (the position-dependent power spectrum), and the collapsed part is the covariance of the power spectrum across subvolumes of the survey, both available from the data at the ten-percent level. The residual is one amplitude per term, fitted to the mock covariance and checked against those data-side estimates.

9 Checks and caveats

- **Box limit.** Both classes reduce to eq. (49) of Sugiyama et al. (2020) when $W = 1$; in the truncated scheme only the $\lambda_1 = \lambda_2 = 0$ components of $\mathcal{Q}$ survive, with the closed form of the status box, so the box limit tests the kernels, the counting factors and the $\mathcal{T}$ coefficients independently of the pair counts (a 4000-realisation Gaussian box reproduced the class-1 and class-2 box formulae for $B_{000}, B_{202}, B_{220}$ during the review).
- **Gaussian-field test.** As for the power spectrum, Gaussian realisations on the real footprint (`run_gaussian_footprint.py`) measured with the Sugiyama estimator give the exact Gaussian covariance of the windowed multipoles, including the window three-point effects at all ranks, and test λ_w .
- **Which momentum carries the power spectrum.** In Equation 12 to Equation 14 the spectrum is evaluated at the primed momentum; the symmetric choice (the mean over the two shells, or half the sum of the two assignments) differs at $O(\Delta k/k)$ and is the one to use for the final product, as for the power spectrum.
- **Beat-mode pairings.** Of the pairings with a pair inside each estimator, eight need a shell momentum below the window scale and vanish; $(12)(1'2')(33')$ does not (status box, item v) and is kept as a seventh Gaussian term on the $(ii; i'i')$ elements. The same beat-mode pairings exist in the PB, BB and PT terms.
- **Scope.** The Gaussian term is derived in full; the PB, BB and PT terms are derived to the point where their window functions (pair counts) and kernels are identified, with the harmonic chain written out for PB only; P_5 and P_6 are dropped; the super-sample term is separate (see the squeezed limit of PT). The window is the local one; a pair-averaged window enters, as for the power spectrum, through the dilution of the mean and through W in Equation 27.